Effects of Sensor Position Errors on Farfield/Nearfield Wideband Beamformers for Microphone Arrays

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***Abstract*— It is well known that the presence of sensor position errors may lead to failure of farfield/nearfield wideband beamformers for microphone arrays. Therefore, many efforts have been made to design wideband beamformers for microphone arrays robust against sensor position errors in recent years. This paper presents the findings obtained on the study and analysis of the effects of sensor position errors to the performance of farfield/nearfield wideband beamformers for microphone arrays. Several useful properties have been derived through the conduct of a comprehensive statistical analysis, including the bias and variance analysis of array response, to provide in-depth insights into the robustness characteristics of farfield and nearfield wideband beamformers in the presence of sensor position errors. The analysis does not assume any specific array geometry and is thus applicable to arrays with arbitrary geometries. Numerical examples obtained via Monte Carlo simulations are presented to verify the proposed theoretical analysis. This paper is helpful to better understand the effects of sensor position errors on wideband beamformers and useful in microphone array design.**

***Index Terms*— Wideband beamformer, sensor position error, microphone array.**

I. Introduction

BEAMFORMING, i.e., spatial filtering, is one of the key technologies in sensor array signal processing [1]. For audio and speech applications, wideband beamforming with microphone arrays has been widely used in practice. The typical applications include hands-free telephony, hearing aids, multimedia, and speech recording in car environments [2]–[4]. To date, various design approaches for microphone array based wideband beamformers have been proposed in the literature [2]–[13]. In general, these beamformers can be categorized, according to the source-to-array distance, as farfield or nearfield beamformers. As a rule of thumb, when the source-to-array distance is less than $2l^2/\lambda$, where l is the largest array dimension and λ is the operating wavelength, then a nearfield beamformer should be applied [10].

The sensor placement affects the performance of beamformers significantly, and is thus an important issue in the design of a microphone array system. In [14], several criteria are presented on how the optimal microphone positions can be determined. In [15], some heuristic optimization approaches and an exhaustive search approach are reported on how the optimal microphone positions can be determined for superdirective beamformers. In [16], a constrained nonlinear optimization technique is used to obtain the optimal microphone positions and beamformer weight vectors simultaneously. In [17], a hybrid stochastic-analytical method is proposed to synthesize a sparse linear array so that it will have a wideband superdirective beampattern that is robust against microphone gain and phase mismatches. More recently, a compressive sensing based optimization approach is proposed for the design of wideband beamformers with sparse microphone placement [18].

In practice, sensor position errors are usually inevitable in sensor array installation [19]–[28] and several methods have been proposed in the literature to mitigate this problem. These include the array calibration approach [19], [22], [26]–[28], the robust weighted least squares approach [21], the worst-case performance optimization approach [12], etc. Although these methods have been shown to be effective in the presence of microphone position errors, to the best of the authors' knowledge, there is a lack of in-depth theoretical analysis on the effects of microphone position errors to the performance of wideband beamformers. Such an analysis will provide a better understanding of the robustness characteristics of microphone arrays. It is noted that, inspired by the sensitivity analysis on antenna arrays [29], [30], the study on the statistical mean of the squared array response magnitude, also known as the mean of the beam power pattern, of farfield wideband beamformers in the presence of microphone position errors has been presented [5], [6]. However, the important statistical properties, namely the bias and variance analysis of array response, is not reported therein. Moreover, the sensitivity analysis reported assumes that a linear array is used and that sensor position errors occur with a Gaussian distribution and along the linear array axis. As a consequence, the robustness analysis is rather limited in scope, and the conclusions drawn therein are only applicable to the associated special scenario, as will be clear in the sequel. Recently, the effect of microphone position

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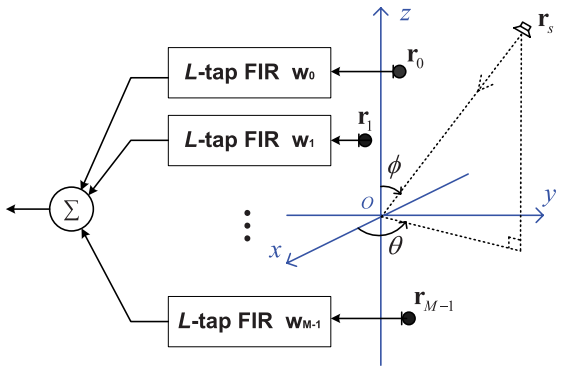


Fig. 1. Farfield/Nearfield wideband beamformer configuration and the definition of the coordinate system.

errors to the performance of farfield wideband beamformers is studied in [31] by performing variance analysis of array response. However, the work has the same limitation that, it assumes the use of a linear array and that microphone position errors occur only along the array axis. Moreover, the statistical analysis only focuses on the array response variance (ARV), and another important statistical property, i.e., the array response bias (ARB), has not been studied yet. Therefore, a comprehensive statistical analysis for farfield wideband beamformers with an arbitrary array needs to be conducted. In addition, studies on the effect of microphone position errors for nearfield wideband beamformers have also not been reported. These are the main motivations for this work.

This paper presents the findings obtained on the conduct of a comprehensive statistical analysis on the effects of sensor position errors to the performance of both farfield and nearfield wideband beamformers for microphone arrays with arbitrary geometries. The rest of the paper is organized as follows. The mathematical data models as well as the definitions of ARB and ARV for farfield and nearfield wideband beamformers are briefly described in Section II. The proposed statistical analysis, the ARB and ARV, and the associated properties derived are presented in Section III and IV respectively. Numerical results obtained via computer simulation are shown in Section V. Finally, Section VI summarizes the main findings and concludes the paper.

II. Background

A. Mathematical Data Model

Consider an M-element microphone array with an arbitrary geometry, where the mth microphone is placed at the nominal position $\mathbf{r}_m = [x_m, y_m, z_m]^T$, $m = 0, \cdots, M-1$, as shown in Fig. 1, where the coordinate system used is also defined. Assume that an L-tap finite impulse response (FIR) filter $\mathbf{w}_m \in \mathbb{R}^{L\times 1}$ is used after the mth microphone. Then the weight vector of a farfield/nearfield wideband beamformer can be defined as $\mathbf{w} = \left[\mathbf{w}_0^T, \mathbf{w}_1^T, \cdots, \mathbf{w}_{M-1}^T\right]^T \in \mathbb{R}^{ML\times 1}$, where the superscript $(\cdot)^T$ denotes the transpose.

1) Farfield Wideband Beamformers: The array response of the farfield wideband beamformer to a source from the angle of arrival (AoA) (θ, ϕ) with the azimuth angle $\theta \in [0, 360°)$ and the elevation angle $\phi \in [0, 180°]$, at frequency f, can be expressed as [7]

$$P(\theta, \phi; f) = \mathbf{w}^T \mathbf{a}(\theta, \phi; f) \tag{1}$$

where $\mathbf{a}(\theta, \phi; f) = \mathbf{h}(\theta, \phi; f) \otimes \mathbf{e}(f)$ is the farfield array response vector, $\otimes$ stands for the Kronecker product, and

$$\mathbf{h}(\theta, \phi; f) = \left[h_0(\theta, \phi; f), h_1(\theta, \phi; f), \cdots, h_{M-1}(\theta, \phi; f)\right]^T \tag{2}$$

$$\mathbf{e}(f) = \left[1, e^{-j2\pi f/f_s}, \cdots, e^{-j2\pi f(L-1)/f_s}\right]^T \tag{3}$$

where $h_m(\theta, \phi; f)$ in (2) is the farfield microphone transfer function from the reference point to the mth microphone, and f_s represents the sampling frequency. Without loss of generality, the coordinate origin has been chosen as the zero phase reference point hereafter. For farfield wideband beamformers without microphone position errors, we have [21]

$$h_m(\theta, \phi; f) = \exp\left\{-j2\pi f \mathbf{u}_s^T \mathbf{r}_m / c\right\} \tag{4}$$

where $\mathbf{u}_s = [\sin\phi\cos\theta, \sin\phi\sin\theta, \cos\phi]^T$ is the direction vector of the source with AoA (θ, ϕ), and c stands for the speed of sound in air.

In the presence of perturbations in microphone positions, the farfield microphone transfer function (4) now becomes[1]

$$\widetilde{h}_m(\theta, \phi; f) = \exp\left\{-j2\pi f(\mathbf{u}_s^T \mathbf{r}_m + \mathbf{u}_s^T \Delta\mathbf{r}_m)/c\right\} \tag{5}$$

where $\Delta\mathbf{r}_m = [\delta_m^x, \delta_m^y, \delta_m^z]^T$ denotes the position error of the mth microphone. In the following, we make the assumption that the microphone position errors are uncorrelated random variables, where the means of δ_m^x, δ_m^y, and δ_m^z are denoted by μ_x, μ_y, and μ_x, respectively, and the variances of δ_m^x, δ_m^y, and δ_m^z are denoted by σ_x^2, σ_y^2, and σ_z^2, respectively.[2]

2) Nearfield Wideband Beamformers: The array response of the nearfield wideband beamformer to a source at location $\mathbf{r}_s = [x_s, y_s, z_s]^T$ with frequency f can be expressed as [12]

$$P(\mathbf{r}_s; f) = \mathbf{w}^T \mathbf{d}(\mathbf{r}_s; f) \tag{6}$$

where $\mathbf{d}(\mathbf{r}_s; f) = \mathbf{h}(\mathbf{r}_s; f) \otimes \mathbf{e}(f)$ is the nearfield array response vector, with

$$\mathbf{h}(\mathbf{r}_s; f) = \left[h_0(\mathbf{r}_s; f), h_1(\mathbf{r}_s; f), \cdots, h_{M-1}(\mathbf{r}_s; f)\right]^T \tag{7}$$

where $h_m(\mathbf{r}_s; f)$ denotes the nearfield microphone transfer function from the source to the mth microphone. The microphone transfer function for nearfield wideband beamformers without microphone position errors can be represented as [11]

$$h_m(\mathbf{r}_s; f) = \frac{\exp\{-j2\pi f\|\mathbf{r}_s - \mathbf{r}_m\|/c\}}{\|\mathbf{r}_s - \mathbf{r}_m\|} \tag{8}$$

where $\|\cdot\|$ denotes the Euclidean norm.

[1]Throughout the paper, the wavy overline $\widetilde{(\cdot)}$ indicates the one with microphone position perturbations.

[2]Unlike [6] and [21], we do not assume the probability density functions of microphone position errors are *a prior* known, thus it may reveal the universal robustness properties of microphone arrays.

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In the presence of perturbations in microphone positions, following (8), the nearfield microphone transfer function can be expressed as

$$\widetilde{h}_m(\mathbf{r}_s;\, f) = \frac{\exp\{-j2\pi f\|\mathbf{r}_s - \mathbf{r}_m - \Delta\mathbf{r}_m\|/c\}}{\|\mathbf{r}_s - \mathbf{r}_m - \Delta\mathbf{r}_m\|}, \tag{9}$$

where $\Delta\mathbf{r}_m$ is the corresponding position error of the mth microphone as defined in (5).

B. Definitions of ARB and ARV

To study the effects of microphone position errors on wideband beamformers from a statistical perspective, hereby we introduce the ARB and ARV of farfield and nearfield wideband beamformers.

The ARB of farfield and nearfield wideband beamformers is defined respectively as

$$\text{Bias}[\widetilde{P}(\theta,\phi;\, f)] \triangleq \mathbb{E}\{\widetilde{P}(\theta,\phi;\, f)\} - P(\theta,\phi;\, f) \tag{10}$$

$$\text{Bias}[\widetilde{P}(\mathbf{r}_s;\, f)] \triangleq \mathbb{E}\{\widetilde{P}(\mathbf{r}_s;\, f)\} - P(\mathbf{r}_s;\, f) \tag{11}$$

where $\mathbb{E}\{\cdot\}$ denotes the mathematical expectation.

The ARV of farfield and nearfield wideband beamformers is defined respectively as

$$\text{Var}[\widetilde{P}(\theta,\phi;\, f)] \triangleq \mathbb{E}\left\{\left|\widetilde{P}(\theta,\phi;\, f) - \mathbb{E}\{\widetilde{P}(\theta,\phi;\, f)\}\right|^2\right\} \tag{12}$$

$$\text{Var}[\widetilde{P}(\mathbf{r}_s;\, f)] \triangleq \mathbb{E}\left\{\left|\widetilde{P}(\mathbf{r}_s;\, f) - \mathbb{E}\{\widetilde{P}(\mathbf{r}_s;\, f)\}\right|^2\right\}. \tag{13}$$

III. The Statistical Analysis on Farfield Wideband Beamformers

In this section, the effects of microphone position errors on farfield wideband beamformers are studied through statistical analysis, and several interesting properties are derived.

A. ARB Analysis

For small microphone position errors, applying Taylor series expansion with first-order approximation to (5) yields

$$\begin{aligned}\widetilde{h}_m(\theta,\phi;\, f) &\simeq \left[1 - j2\pi f\mathbf{u}_s^T\,\Delta\mathbf{r}_m/c\right]\exp\left\{-j2\pi f\mathbf{u}_s^T\,\mathbf{r}_m/c\right\}\\ &\triangleq h_m(\theta,\phi;\, f) + \Delta h_m(\theta,\phi;\, f)\end{aligned} \tag{14}$$

where

$$\Delta h_m(\theta,\phi;\, f) = \left[-j2\pi f\mathbf{u}_s^T\,\Delta\mathbf{r}_m/c\right]\exp\left\{-j2\pi f\mathbf{u}_s^T\,\mathbf{r}_m/c\right\} \tag{15}$$

is the perturbation of microphone transfer function due to microphone position errors.

Using (14), we can reformulate (1) as

$$\widetilde{P}(\theta,\phi;\, f) = \mathbf{w}^T\,[\mathbf{a}(\theta,\phi;\, f) + \Delta\mathbf{a}(\theta,\phi;\, f)] \tag{16}$$

where

$$\mathbf{a}(\theta,\phi;\, f) = \mathbf{h}(\theta,\phi;\, f)\otimes\mathbf{e}(f) \tag{17}$$

$$\Delta\mathbf{a}(\theta,\phi;\, f) = \Delta\mathbf{h}(\theta,\phi;\, f)\otimes\mathbf{e}(f) \tag{18}$$

with

$$\begin{aligned}\Delta\mathbf{h}(\theta,\phi;\, f) = [&\Delta h_0(\theta,\phi;\, f),\, \Delta h_1(\theta,\phi;\, f),\\ &\cdots,\, \Delta h_{M-1}(\theta,\phi;\, f)]^T.\end{aligned} \tag{19}$$

Theorem 1: The magnitude of ARB of a farfield wideband beamformer in the presence of microphone position errors is given by

$$\left|\text{Bias}[\widetilde{P}(\theta,\phi;\, f)]\right| \simeq (2\pi f/c)\left|\mathbf{u}_s^T\boldsymbol{\mu}\right|\cdot|P(\theta,\phi;\, f)| \tag{20}$$

where $\boldsymbol{\mu} = [\mu_x, \mu_y, \mu_z]^T$.

Proof: Using (16) and (1), we have

$$\begin{aligned}&\left|\text{Bias}[\widetilde{P}(\theta,\phi;\, f)]\right|\\ &\quad= \left|\mathbb{E}\{\widetilde{P}(\theta,\phi;\, f)\} - P(\theta,\phi;\, f)\right|\\ &\quad= \left|\mathbb{E}[\mathbf{w}^T\,\Delta\mathbf{a}(\theta,\phi;\, f)]\right|\\ &\quad= \left|\mathbf{w}^T\,\{\mathbb{E}[\Delta\mathbf{h}(\theta,\phi;\, f)]\otimes\mathbf{e}(f)\}\right|\\ &\quad\simeq (2\pi f/c)\left|\mathbf{u}_s^T\boldsymbol{\mu}\right|\cdot\left|\mathbf{w}^T\,[\mathbf{h}(\theta;\phi;\, f)\otimes\mathbf{e}(f)]\right|\\ &\quad= (2\pi f/c)\left|\mathbf{u}_s^T\boldsymbol{\mu}\right|\cdot|P(\theta,\phi;\, f)|.\end{aligned}$$

■

With Theorem 1, the following findings are in order.

Finding 1: The ARB of a farfield wideband beamformer is frequency dependent. Interestingly, for the widely used frequency-invariant farfield beamformers (FIFB) in microphone arrays [9], the magnitude of ARB will increase with frequency increasing, i.e., the FIFB will be more biased in high frequencies in the presence of microphone position errors.

Finding 2: The ARB of a farfield wideband beamfomer is dependent of the source AoA (θ,ϕ). In particular, for omnidirectional perturbations[3] in microphone positions, the ARB tends to zero, i.e., the array response is nearly unbiased, when the source AoA is around $(\arccos(-\cot\phi/\sqrt{2}) + 45°, \phi)$ and $(\arccos(-\cot\phi/\sqrt{2}) - 135°, \phi)$ if $35.26° \le \phi \le 45°$, or around $(\arccos(-\cot\phi/\sqrt{2}) + 45°, \phi)$ and $(\arccos(-\cot\phi/\sqrt{2}) + 225°, \phi)$ if $45° < \phi \le 144.73°$ (See Appendix A for a proof). It is worth noting that these unbiased array responses just can be achievable within the limited space away from the z-axis. For unidirectional microphone position perturbation,[4] the magnitude of ARB of a farfield wideband beamformer is proportional to the magnitude of the cosine of the angle between the direction vector of the source and the direction vector of microphone position perturbations.

Finding 3: It is known that the speed of sound will not keep constant in practice due to temperature variation. When the medium is homogeneous and at rest, it holds that $c = 331.4 + 0.6\rho$ m/s, where ρ represents the temperature in centigrade [32]. From Theorem 1, it follows that the magnitude of the relative bias of array response, $2\pi f\left|\mathbf{u}_s^T\boldsymbol{\mu}\right|/c$, is inversely proportional to the speed of sound or the temperature in homogeneous environment. Specifically,

[3]By omnidirectional, it refers to the case that the statistical characteristics (means and variances) of the microphone position perturbations along the x-, y-, and z-axis are the same. Mathematically, it implies that $\mu_x = \mu_y = \mu_z$ and $\sigma_x^2 = \sigma_y^2 = \sigma_z^2$.

[4]By unidirectional, it means that the perturbation directions of all microphones are along the same direction. Mathematically, it can be defined as follows. Denote the direction vectors of position perturbations of the mth microphone and the m'th microphone respectively as $\mathbf{a}_m = [\sin\phi_m\cos\theta_m, \sin\phi_m\sin\theta_m, \cos\phi_m]^T$ and $\mathbf{a}_{m'} = [\sin\phi_{m'}\cos\theta_{m'}, \sin\phi_{m'}\sin\theta_{m'}, \cos\phi_{m'}]^T$, $(m, m' = 0, \ldots, M-1)$. Then it implies that $\mathbf{a}_m = \mathbf{a}_{m'}$ or $\mathbf{a}_m = -\mathbf{a}_{m'}$.

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when the nominal array response $P(\theta, \phi; f)$ is relatively insensitive to variation of the speed of sound,[5] it can be concluded that the array response will be more biased if the speed of sound is decreased, i.e., with the temperature decreasing in homogeneous environment.

B. ARV Analysis

Now we perform the ARV analysis for farfield wideband beamformers. First, we introduce the following theorem.

Theorem 2: The ARV of a farfield wideband beamformer in the presence of microphone position errors is given by

$$\begin{aligned}\mathrm{Var}[\widetilde{P}(\theta,\phi;f)] &\simeq (2\pi f/c)^2[\sigma_x^2(\sin\phi\cos\theta)^2+\sigma_y^2(\sin\phi\sin\theta)^2 \\ &\quad+\sigma_z^2(\cos\phi)^2]\mathbf{w}^T[\mathbf{I}_M\otimes\Lambda(f)]\mathbf{w} \qquad (21)\end{aligned}$$

where $\mathbf{I}_M$ denotes the $M \times M$ identity matrix, and $\Lambda(f)$ is the $L \times L$ symmetric matrix whose (k, l)th element is $\Lambda^{(kl)}(f) = \cos[2\pi(k-l)f/f_s]$.

Proof: See Appendix B. ∎

With Theorem 2, regarding the effects of microphone position errors on the ARV of farfield wideband beamformers, we have the following findings.

Finding 4: For omnidirectional perturbations in microphone positions, the source AoA has little effect on the ARV of farfield wideband beamformers.[6] This is unlike the effect on the ARB, which is dependent of the source AoA as discussed in Finding 2. Therefore, the robustness of farfield wideband beamformers with omnidirectional microphone position perturbations will also depend on source AoA. Specifically, when the source AoA is around the directions specified in Finding 2, the array response of farfield wideband beamformers will be more statistically robust.

Finding 5: For unidirectional perturbations in microphone positions, the ARV of a farfield wideband beamformer is proportional to the square of the cosine of the angle between the source direction vector and the direction vector of microphone perturbations.[7] It follows that the ARV of a farfield wideband beamformer is dependent of source AoA in the presence of unidirectional perturbations in microphone positions. Moreover, the closer the direction of the incident source is to the perturbation direction of microphone positions, the larger the ARV will be, which is consistent well with the effect on the ARB, as revealed in Finding 2. In particular, the farfield wideband beamformers will be more statistically robust when the direction of incident source is perpendicular to the perturbation direction of microphone positions, while it will become less robust when the direction of incident source is parallel to the perturbation direction of microphone positions.

[5]It has been found by extensive numerical evaluations that some typical farfield wideband beamformers in the literature may possess this characteristic, see [8] and [9].

[6]For omnidirectional perturbations, we have $\sigma_x^2 = \sigma_y^2 = \sigma_z^2 \triangleq \sigma^2$. Then, by (21) it follows that $\mathrm{Var}[\widetilde{P}(\theta,\phi;f)] = (2\pi f/c)^2\sigma^2\mathbf{w}^T[\mathbf{I}_M\otimes\Lambda(f)]\mathbf{w}$.

[7]Denote the direction vector of microphone perturbation as $\mathbf{a} = [\sin\phi'\cos\theta', \sin\phi'\sin\theta', \cos\phi']^T$. It holds that $\mathrm{Var}[\widetilde{P}(\theta,\phi;f)] \simeq (2\pi f/c)^2\sigma^2(\mathbf{u}_s^T\mathbf{a})^2\mathbf{w}^T[\mathbf{I}_M\otimes\Lambda(f)]\mathbf{w} = (2\pi f\cos\psi/c)^2\sigma^2\mathbf{w}^T[\mathbf{I}_M\otimes\Lambda(f)]\mathbf{w}$, where ψ is the angle between $\mathbf{a}$ and $\mathbf{u}_s$, and σ^2 denotes the variance of the microphone perturbation.

For widely used linear microphone arrays, if the microphone position errors are along the axis of linear arrays, i.e., unidirectional microphone perturbations, it follows directly from Finding 5 that the robustness of farfield beamformers, due to the effect of microphone position errors, will degrade when the source moves away from the broadside direction of linear arrays. Interestingly, this agrees well with the property derived in [6], though the two studies are from different perspectives. It is noted that, the sensitivity analysis of microphone position errors [6] is based on an *ad hoc* perturbation model for linear arrays. From our analysis, it shows that the sensitivity property shown is only applicable to that particular perturbation model, and no longer hold if the microphone perturbation direction is perpendicular to the array axis or in the case of omnidirectional microphone perturbations.

Finding 6: The ARV of a farfield wideband beamformer is inversely proportional to the square of the speed of sound. Moreover, recall the effect of microphone position errors on the ARB of farfield wideband beamformers, we can arrive at the conclusion that farfield wideband beamformers are more robust at higher temperatures than at lower temperatures in the presence of microphone position errors, when the nominal array response is relatively insensitive to sound speed variation in homogeneous environments.

IV. The Statistical Analysis on Nearfield Wideband Beamformers

In this section, the statistical analysis of nearfield wideband beamformers is presented in the presence of microphone position errors, and following the theoretical studies several insightful properties are revealed.

A. ARB Analysis

For small microphone position errors, applying Taylor series expansion with first-order approximation to (9) yields

$$\begin{aligned}\widetilde{h}_m(\mathbf{r}_s;f) &\simeq \left\{1+\Delta\mathbf{r}_m^T(\mathbf{r}_s-\mathbf{r}_m)\left(\frac{1}{\|\mathbf{r}_s-\mathbf{r}_m\|^2}+j\frac{2\pi f}{c\|\mathbf{r}_s-\mathbf{r}_m\|}\right)\right\} \\ &\quad\times\frac{\exp\{-j2\pi f\|\mathbf{r}_s-\mathbf{r}_m\|/c\}}{\|\mathbf{r}_s-\mathbf{r}_m\|} \\ &\triangleq h_m(\mathbf{r}_s;f)+\Delta h_m(\mathbf{r}_s;f) \qquad (22)\end{aligned}$$

where

$$\begin{aligned}\Delta h_m(\mathbf{r}_s;f) &= \Delta\mathbf{r}_m^T(\mathbf{r}_s-\mathbf{r}_m)\left(\frac{1}{\|\mathbf{r}_s-\mathbf{r}_m\|}+j\frac{2\pi f}{c}\right) \\ &\quad\times\frac{\exp\{-j2\pi f\|\mathbf{r}_s-\mathbf{r}_m\|/c\}}{\|\mathbf{r}_s-\mathbf{r}_m\|^2} \qquad (23)\end{aligned}$$

is the perturbation of microphone transfer function due to microphone position errors.

With the help of (22), we can reformulate (6) as

$$\widetilde{P}(\mathbf{r}_s;f) = \mathbf{w}^T[\mathbf{d}(\mathbf{r}_s;f)+\Delta\mathbf{d}(\mathbf{r}_s;f)] \qquad (24)$$

where

$$\mathbf{d}(\mathbf{r}_s;f) = \mathbf{h}(\mathbf{r}_s;f)\otimes\mathbf{e}(f) \qquad (25)$$

$$\Delta\mathbf{d}(\mathbf{r}_s;f) = \Delta\mathbf{h}(\mathbf{r}_s;f)\otimes\mathbf{e}(f) \qquad (26)$$

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with

$$\Delta\mathbf{h}(\mathbf{r}_s;\, f)=\left[\Delta h_0(\mathbf{r}_s;\, f),\, \Delta h_1(\mathbf{r}_s;\, f),\, \cdots,\, \Delta h_{M-1}(\mathbf{r}_s;\, f)\right]^T. \tag{27}$$

Theorem 3: The ARB of a nearfield wideband beamformer with microphone position errors satisfies

$$\left|\text{Bias}[\widetilde{P}(\mathbf{r}_s;\, f)]\right| \simeq \Big|\mathbf{w}^T\left\{[\mu_x\mathbf{b}_x(\mathbf{r}_s;\, f)+\mu_y\mathbf{b}_y(\mathbf{r}_s;\, f) + \mu_z\mathbf{b}_z(\mathbf{r}_s;\, f)]\otimes\mathbf{e}(f)\right\}\Big| \tag{28}$$

where $\mathbf{b}_x(\mathbf{r}_s;\, f) = [b_{0,x}(\mathbf{r}_s;\, f), b_{1,x}(\mathbf{r}_s;\, f), \ldots, b_{M-1,x}(\mathbf{r}_s;\, f)]^T$ with

$$b_{m,x}(\mathbf{r}_s;\, f)=h_m(\mathbf{r}_s;\, f)\left[\frac{x_s-x_m}{\|\mathbf{r}_s-\mathbf{r}_m\|^2}+j(2\pi f/c)\frac{x_s-x_m}{\|\mathbf{r}_s-\mathbf{r}_m\|}\right],$$

$\mathbf{b}_y(\mathbf{r}_s;\, f) = [b_{0,y}(\mathbf{r}_s;\, f), b_{1,y}(\mathbf{r}_s;\, f), \ldots, b_{M-1,y}(\mathbf{r}_s;\, f)]^T$ with

$$b_{m,y}(\mathbf{r}_s;\, f)=h_m(\mathbf{r}_s;\, f)\left[\frac{y_s-y_m}{\|\mathbf{r}_s-\mathbf{r}_m\|^2}+j(2\pi f/c)\frac{y_s-y_m}{\|\mathbf{r}_s-\mathbf{r}_m\|}\right],$$

and $\mathbf{b}_z(\mathbf{r}_s;\, f) = [b_{0,z}(\mathbf{r}_s;\, f), b_{1,z}(\mathbf{r}_s;\, f), \ldots, b_{M-1,z}(\mathbf{r}_s;\, f)]^T$ with

$$b_{m,z}(\mathbf{r}_s;\, f)=h_m(\mathbf{r}_s;\, f)\left[\frac{z_s-z_m}{\|\mathbf{r}_s-\mathbf{r}_m\|^2}+j(2\pi f/c)\frac{z_s-z_m}{\|\mathbf{r}_s-\mathbf{r}_m\|}\right].$$

Regarding the property of the ARB for nearfield wideband beamformers, we have the following proposition.

Proposition 1: It holds that $\text{Bias}[\widetilde{P}(\mathbf{r}_s;\, f)] \to 0$, when $\|\mathbf{r}_s-\mathbf{r}_m\|\to\infty$.

Proof: From (28), we have

$$\begin{aligned}
&\left|\text{Bias}[\widetilde{P}(\mathbf{r}_s;\, f)]\right| \\
&\simeq \left|\mathbf{w}^T\left\{[\mu_x\mathbf{b}_x(\mathbf{r}_s;\, f)+\mu_y\mathbf{b}_y(\mathbf{r}_s;\, f)+\mu_z\mathbf{b}_z(\mathbf{r}_s;\, f)]\otimes\mathbf{e}(f)\right\}\right| \\
&\le \|\mathbf{w}\|\cdot\|[\mu_x\mathbf{b}_x(\mathbf{r}_s;\, f)+\mu_y\mathbf{b}_y(\mathbf{r}_s;\, f)+\mu_z\mathbf{b}_z(\mathbf{r}_s;\, f)]\otimes\mathbf{e}(f)\| \\
&\le |\mu_x|\,\|\mathbf{w}\| \\
&\quad\times\sqrt{L\sum_{m=0}^{M-1}\frac{(x_s-x_m)^2}{\|\mathbf{r}_s-\mathbf{r}_m\|^4}\left[\frac{1}{\|\mathbf{r}_s-\mathbf{r}_m\|^2}+\left(\frac{2\pi f}{c}\right)^2\right]} \\
&\quad+\left|\mu_y\right|\|\mathbf{w}\| \\
&\quad\times\sqrt{L\sum_{m=0}^{M-1}\frac{(y_s-y_m)^2}{\|\mathbf{r}_s-\mathbf{r}_m\|^4}\left[\frac{1}{\|\mathbf{r}_s-\mathbf{r}_m\|^2}+\left(\frac{2\pi f}{c}\right)^2\right]} \\
&\quad+|\mu_z|\,\|\mathbf{w}\| \\
&\quad\times\sqrt{L\sum_{m=0}^{M-1}\frac{(z_s-z_m)^2}{\|\mathbf{r}_s-\mathbf{r}_m\|^4}\left[\frac{1}{\|\mathbf{r}_s-\mathbf{r}_m\|^2}+\left(\frac{2\pi f}{c}\right)^2\right]} \\
&\le (|\mu_x|+\left|\mu_y\right|+|\mu_z|)\|\mathbf{w}\| \\
&\quad\times\sqrt{L\sum_{m=0}^{M-1}\left[\frac{1}{\|\mathbf{r}_s-\mathbf{r}_m\|^4}+\frac{(2\pi f/c)^2}{\|\mathbf{r}_s-\mathbf{r}_m\|^2}\right]}.
\end{aligned} \tag{29}$$

Notice that the right hand side of (29) tends to zero when $\|\mathbf{r}_s-\mathbf{r}_m\|\to\infty$. Therefore, it follows that $\text{Bias}[P(\mathbf{r}_s;\, f)]\to 0$ when $\|\mathbf{r}_s-\mathbf{r}_m\|\to\infty$. ■

Finding 7: From the above proposition, we can conclude that the array response of nearfield wideband beamformers will tend toward being unbiased, i.e., close to the nominal array response, as the source-to-array distance increases. Notice that this property holds regardless of unidirectional or omnidirectional perturbations in microphone positions.

B. ARV Analysis

The ARV of nearfield wideband beamformers is summarized in the following theorem.

Theorem 4: The ARV of a nearfield wideband beamformer with microphone position errors is given by

$$\text{Var}[\widetilde{P}(\mathbf{r}_s;\, f)] \simeq \mathbf{w}^T\{[\sigma_x^2\Phi_x(\mathbf{r}_s;\, f)+\sigma_y^2\Phi_y(\mathbf{r}_s;\, f) + \sigma_z^2\Phi_z(\mathbf{r}_s;\, f)]\otimes\Lambda(f)\}\mathbf{w} \tag{30}$$

where the matrices $\Phi_x(\mathbf{r}_s;\, f)$, $\Phi_y(\mathbf{r}_s;\, f)$, and $\Phi_z(\mathbf{r}_s;\, f)$ are all $M\times M$ diagonal matrices, whose (m,m)th elements are respectively given by

$$\Phi_x^{(mm)}(\mathbf{r}_s;\, f)=\frac{(x_s-x_m)^2}{\|\mathbf{r}_s-\mathbf{r}_m\|^4}\left[\frac{1}{\|\mathbf{r}_s-\mathbf{r}_m\|^2}+\left(\frac{2\pi f}{c}\right)^2\right] \tag{31}$$

$$\Phi_y^{(mm)}(\mathbf{r}_s;\, f)=\frac{(y_s-y_m)^2}{\|\mathbf{r}_s-\mathbf{r}_m\|^4}\left[\frac{1}{\|\mathbf{r}_s-\mathbf{r}_m\|^2}+\left(\frac{2\pi f}{c}\right)^2\right] \tag{32}$$

$$\Phi_z^{(mm)}(\mathbf{r}_s;\, f)=\frac{(z_s-z_m)^2}{\|\mathbf{r}_s-\mathbf{r}_m\|^4}\left[\frac{1}{\|\mathbf{r}_s-\mathbf{r}_m\|^2}+\left(\frac{2\pi f}{c}\right)^2\right]. \tag{33}$$

Proof: It can be easily proved following the similar procedures as Theorem 2. ■

From Theorem 4, regarding the effects of microphone position errors on nearfield wideband beamformers, we have the following propositions and findings.

Proposition 2: Consider the omnidirectional perturbation in microphone positions. Given two source positions $\mathbf{r}_s'$ and $\mathbf{r}_s$, with $\|\mathbf{r}_s'-\mathbf{r}_m\| < \|\mathbf{r}_s-\mathbf{r}_m\|$, it holds that $\text{Var}[\widetilde{P}(\mathbf{r}_s',\, f)] \ge \text{Var}[\widetilde{P}(\mathbf{r}_s,\, f)]$.

Proof: Since $\sigma_x^2=\sigma_y^2=\sigma_z^2\triangleq\sigma^2$, from (30) we have

$$\begin{aligned}
\text{Var}[\widetilde{P}(\mathbf{r}_s;\, f)] &\simeq \mathbf{w}^T\{[\sigma_x^2\Phi_x(\mathbf{r}_s;\, f)+\sigma_y^2\Phi_y(\mathbf{r}_s;\, f) \\
&\quad+\sigma_z^2\Phi_z(\mathbf{r}_s;\, f)]\otimes\Lambda(f)\}\mathbf{w} \\
&= \sigma^2\mathbf{w}^T\left[\Phi(\mathbf{r}_s;\, f)\otimes\Lambda(f)\right]\mathbf{w}
\end{aligned} \tag{34}$$

where $\Phi(\mathbf{r}_s;\, f)$ is an $M\times M$ diagonal matrix, whose (m,m)th element is equal to

$$\Phi^{(mm)}(\mathbf{r}_s;\, f)=\frac{1}{\|\mathbf{r}_s-\mathbf{r}_m\|^4}+\frac{4\pi^2 f^2}{c^2\|\mathbf{r}_s-\mathbf{r}_m\|^2}. \tag{35}$$

Considering $\|\mathbf{r}_s'-\mathbf{r}_m\| < \|\mathbf{r}_s-\mathbf{r}_m\|$, we have $\|\mathbf{r}_s'-\mathbf{r}_m\|^{-2} > \|\mathbf{r}_s-\mathbf{r}_m\|^{-2}$. Therefore, it follows that $\Phi^{(mm)}(\mathbf{r}_s';\, f) > \Phi^{(mm)}(\mathbf{r}_s;\, f)$. This implies that $\Phi(\mathbf{r}_s';\, f)-\Phi(\mathbf{r}_s;\, f)$ is positive definite. Reconsider that $\Lambda(f)$ is positive semi-definite, we have $\text{Var}[\widetilde{P}(\mathbf{r}_s',\, f)]-\text{Var}[\widetilde{P}(\mathbf{r}_s,\, f)] \simeq \sigma^2\mathbf{w}^T\{[\Phi(\mathbf{r}_s';\, f)-\Phi(\mathbf{r}_s;\, f)]\otimes\Lambda(f)\}\mathbf{w}\ge 0$. ■

Finding 8: From Proposition 2, it leads to the fact that the ARV of nearfield wideband beamformers will decrease due to omnidirectional perturbations in microphone positions when the source-to-array distance is increasing. This is similar to the effects of microphone gain and phase errors on nearfield wideband beamformers [13]. Moreover, recall from Proposition 1 that the array response of a nearfield wideband beamformer will tend to be unbiased with the source-to-array distance

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$$X_m = \sqrt{\frac{\sqrt{[(y_s - y_m)^2 + (z_s - z_m)^2](2\pi f/c)^2\{[(y_s - y_m)^2 + (z_s - z_m)^2](2\pi f/c)^2 + 1\} + 1} - 1}{(2\pi f/c)^2}}. \quad (36)$$

increasing, we can conclude that the robustness of nearfield beamformers tends to be improved when the source is moving away from microphone arrays. Interestingly, this agrees well with the previous results that are stated in [12, Remark 3], which are derived from a different perspective. Accordingly, the source-to-array distance should be carefully considered in practice for the design of nearfield microphone arrays.

In order to find more insightful properties of nearfield wideband beamformers with microphone position errors, we hereby introduce the following proposition through examining unidirectional perturbations in microphone positions.

Proposition 3: In the presence of the x-axis perturbations in microphone positions, it holds that[8]

1) If $x_1 \le x_2 \le \Delta x_{\min}$, or $x_1 \ge x_2 \ge \Delta x_{\max}$, then $\mathrm{Var}[\widetilde{P}(x_1, y_s, z_s; f)] \le \mathrm{Var}[\widetilde{P}(x_2, y_s, z_s; f)]$;
2) If $y_1 \le y_2 \le y_{\min}$, or $y_1 \ge y_2 \ge y_{\max}$, then $\mathrm{Var}[\widetilde{P}(x_s, y_1, z_s; f)] \le \mathrm{Var}[\widetilde{P}(x_s, y_2, z_s; f)]$;
3) If $z_1 \le z_2 \le z_{\min}$, or $z_1 \ge z_2 \ge z_{\max}$, then $\mathrm{Var}[\widetilde{P}(x_s, y_s, z_1; f)] \le \mathrm{Var}[\widetilde{P}(x_s, y_s, z_2; f)]$;

where $\Delta x_{\min} = \min\{x_m - X_m\}$, $\Delta x_{\max} = \max\{x_m + X_m\}$, $x_{\min} = \min\{x_m\}$, $x_{\max} = \max\{x_m\}$, $(m = 0, 1, \cdots, M-1)$, and X_m are defined at the top of the page.

Proof: See Appendix C. ∎

Based on Proposition 3, the following findings are in order.[9]

Finding 9: In general, the ARV of nearfield wideband beamformers in the presence of unidirectional microphone position errors is dependent of source location. Moreover, outside an uncertainty region whose boundaries depend on the microphone locations, or both source and microphone locations, the ARV of nearfield wideband beamformers will decrease when the source-to-array distance increases. By uncertainty, here we mean that the relationship between the ARV and source-to-array distance is theoretically uncertain. Reconsidering the property of ARB, it follows that the robustness of nearfield microphone arrays will accordingly tend to be improved with the source-to-array distance increasing outside the uncertainty region. Inside the uncertainty region, however, it is unclear in theory how the ARV varies with the source-to-array distance. Regarding the uncertainty region, the following observations can be made:

- When the perturbation direction of microphone positions is perpendicular to source direction, the boundaries of the uncertainty region are dependent of the microphone locations. Interestingly, the size of the uncertainty region is actually equal to the dimension of the microphone array along the impinging direction of the source. Specifically, for the widely used linear or circular microphone arrays, there will be no such uncertainty region if the perturbation direction of microphone positions is along the axis of linear arrays, or lies in the plane of circular arrays.
- When the perturbation direction of microphone positions is parallel to source direction, the boundaries of the uncertainty region will depend on both the microphone locations and the source location. To see this more clearly, let us consider the case with x-axis perturbations in microphone positions. According to Proposition 3.1, the corresponding uncertainty region is given by $(\Delta x_{\min}, \Delta x_{\max})$. For high frequencies, since $2\pi f/c \gg 1$, we have $X_m \simeq \sqrt{(y_s - y_m)^2 + (z_s - z_m)^2}$, which implies that X_m is approximately equal to the length of the line segment which is the projection of the line segment linking the source point and the position of the mth microphone onto the yoz plane. Reconsider that $\Delta x_{\min} \ge x_{\min} - \max\{X_m\}$ and $\Delta x_{\max} \le x_{\max} + \max\{X_m\}$. Then the uncertainty region $(\Delta x_{\min}, \Delta x_{\max})$ can be further approximated by $(x_{\min} - \max\{X_m\}, x_{\max} + \max\{X_m\})$. By performing the y- or z-axis perturbation analysis, similar results can also be obtained. To summarize, the size of the uncertainty region is approximately equal to the summation of two terms. One is the dimension of microphone array along the source direction, and the other is twice of the maximum length of line segments which are the projections of the line segments linking the source point and the positions of all microphones onto the plane that is perpendicular to source direction.

Unlike the effect of unidirectional microphone perturbations, recall from Proposition 2 that there is no such uncertainty region for the case of omnidirectional microphone perturbations.

Finding 10: Following Theorem 4, it holds that the ARV of a nearfield wideband beamformer will increase with the speed of sound decreasing, i.e., corresponding to the temperature increasing in homogeneous environments. This effect of microphone position errors on nearfield wideband beamformers is similar to that on farfield wideband beamformers discussed in Section III. It is interesting to point out that, in contrast, this is also quite different from the effect of microphone gain and phase errors on nearfield wideband beamformers, where the speed of sound has little impact [13].

V. Numerical Evaluation

In this section, simulation results are presented to verify the theoretical analysis. For farfield wideband beamformers, a robust FIFB using Tikhonov regularization [9] is considered, while the nearfield wideband beamformers used are the worst-case performance optimization based method [12].

[8]Only the x-axis perturbations are considered herein due to the space limit. However, the results can be generalized easily to the y- and z-axis perturbations, and similar conclusions can also be derived.

[9]We would like to point out that, although the properties are derived based on the perturbation analysis along the coordinate axis, it has been found that the conclusions drawn herein do hold for the general case with an arbitrary perturbation direction as well. This can be proved through performing rotation of the coordinate system and then by using Proposition 3.

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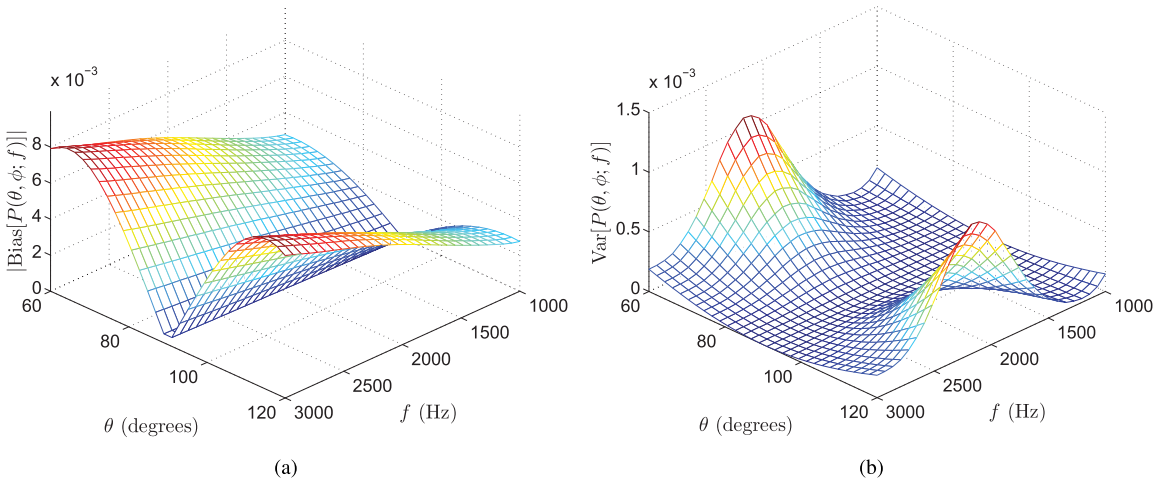


Fig. 2. The ARB and ARV of farfield wideband beamformers with different source AoAs in the presence of unidirectional perturbations in microphone positions. (a) The ARB. (b) The ARV.

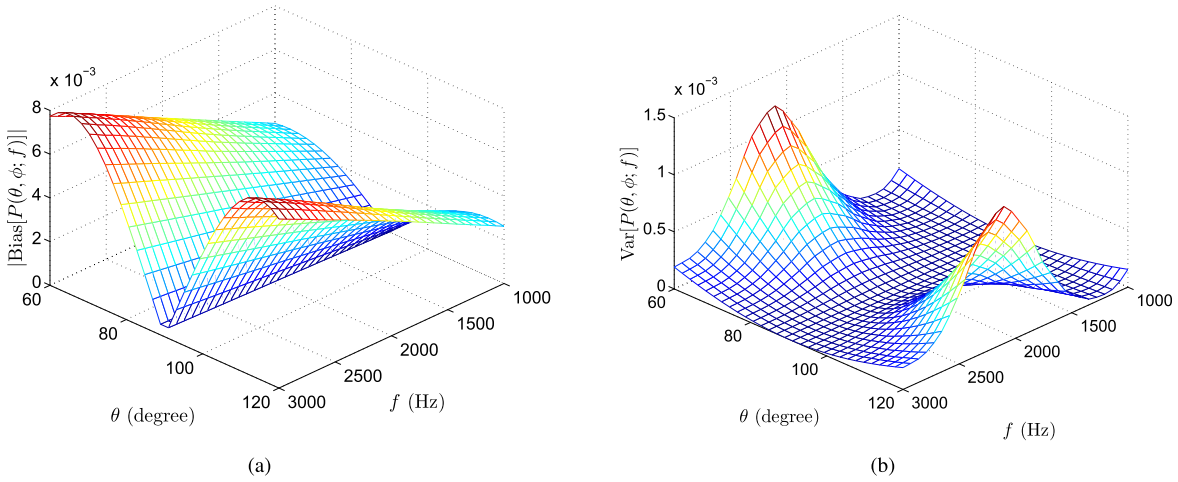


Fig. 3. The theoretical ARB and ARV of farfield wideband beamformers with different source AoAs in the presence of unidirectional perturbations in microphone positions. (a) The theoretical ARB, obtained by using (20). (b) The theoretical ARV, obtained by using (21).

A. Farfield Wideband Beamformers

Consider a seven-element uniform linear array in farfield, which is located in the x-axis with inter-element spacing $d = 4$ cm. The nominal position of the mth microphone is $\mathbf{r}_m = [(m-3)d, 0, 0]^T$, $m = 0, \cdots, 6$. The FIR tap length is chosen as 30, and the sampling frequency is set to 8 kHz. The speed of sound is set to 340 m/s, unless otherwise stated. The passband and stopband regions are defined as $[60 \sim 120°]$ and $[0 \sim 50°] \cup [130 \sim 180°]$, respectively. The frequency band of interest is set to $1000 \sim 3000$ Hz. In the FIFB design, the desired passband response is defined as the mainlobe shape of a Chebyshev 20-dB beampattern. For ease of comparison, the farfield beamformer weights have been normalized such that the peak of the mean array response magnitude is equal to unity. All the results shown below are obtained through 5000 Monte Carlo trials over a specified random set of microphone position perturbations.

1) Unidirectional Perturbations in Microphone Positions: Here, we consider that the perturbation of microphone positions is along the x-axis, where the position errors of all the microphones are uniformly distributed over $[-1, 2]$ mm.[10] Figs. 2(a) and 2(b) show the ARB and ARV of the farfield wideband beamformer with different azimuth angles, respectively, where $\phi = 90°$. As can be seen, the ARB and ARV of the farfield beamformer both increase when the source AoA is away from the array broadside direction. This is well-consistent with the theoretical results, i.e., when the source AoA is close to the microphone perturbation direction, the x-axis, both the ARB and ARV will increase, which indicates that the robustness of the microphone array will degrade. Moreover, as can be seen from Fig. 2(a), since the farfield wideband beamformer is frequency invariant in passband, its ARB in the passband will increase with frequency increasing, as revealed by our theoretical analysis. For comparison, the theoretical ARB and ARV by using (20) and (21) are also shown in Figs. 3(a) and 3(b), respectively. Compared with the

[10]Note here that we are considering the case where microphone position errors are biased with the non-zero mean of 0.5 mm. Otherwise, for unbiased microphone position errors, by (20) it follows that the array response will also be unbiased.

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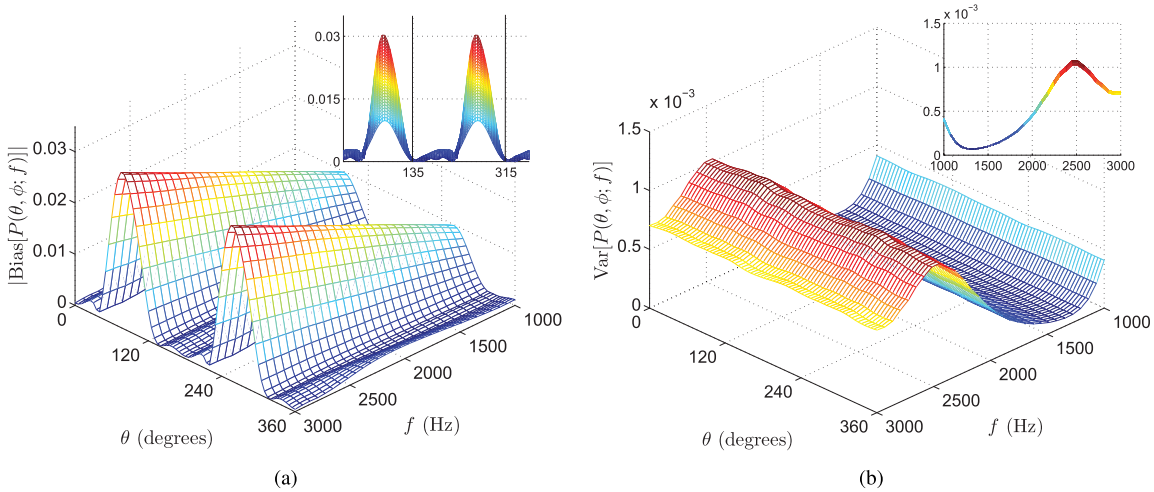


Fig. 4. The ARB and ARV of the farfield wideband beamformer with different source AoAs in the presence of the omnidirectional perturbations in microphone positions. (a) The ARB. (b) The ARV. The side views are shown in the insets.

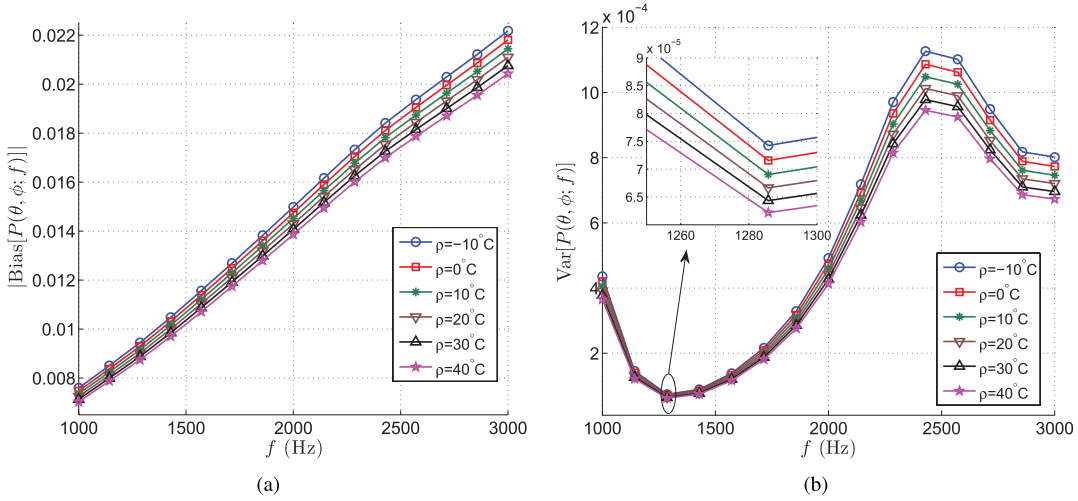


Fig. 5. The ARB and ARV of the farfield wideband beamformer with different temperatures in homogeneous environments. (a) The ARB. (b) The ARV, where a close-up view in lower frequency band is shown in the inset.

simulation results in Figs. 2(a) and 2(b), it can be seen that although an approximation of farfield array model has been used in our statistical analysis, the resultant theoretical results are quite consistent with the simulation results. This is also the case for the following numerical examples. Due to the space limit, we will not present the theoretical results therein. Nonetheless, it suffices to use the simulation results to support the above findings on farfield beamformers.

2) Omnidirectional Perturbations in Microphone Positions: In this simulation, we suppose that the x-, y-, and z-axis perturbations in microphone positions all have a uniform distribution over $[-1, 2]$ mm. Figs. 4(a) and 4(b) display the ARB and ARV of the farfield wideband beamformer with various azimuth angles, respectively, where $\phi = 90°$. According to Finding 2, the array response will be nearly unbiased around $\theta = 135°$ or $315°$, which is confirmed by the simulation results shown in Fig. 4(a). In contrast, the source AoA has little effect on the ARV of the farfield wideband beamformer in the presence of omnidirectional perturbations in microphone positions, as shown in Fig. 4(b). Moreover, similar to the case of unidirectional position perturbations, the ARB of the FIFB in the passband will also increase when the frequency is increasing.

3) Effect of Temperature in Homogeneous Environment: Consider the omnidirectional perturbations in microphone positions, where the x-, y-, and z-axis perturbations are all uniformly distributed over $[-1, 2]$ mm. Here, the source AoA is set to be $\theta = 100°$ and $\phi = 90°$. The ARB and ARV of the FIFB with the temperature from -10 to 40 °C, corresponding to the speed of sound from 325.4 to 355.4 m/s, are plotted in Figs. 5(a) and 5(b), respectively. It has been found through extensive simulations that the nominal array response of the FIFB is relatively insensitive to sound speed variation. As can be seen from Fig. 5, the ARB and ARV of the farfield beamformer both will increase with the temperature decreasing, which agrees well with our theoretical analysis. This suggests that it is better to use microphone arrays at a relatively higher temperature in homogeneous environments in

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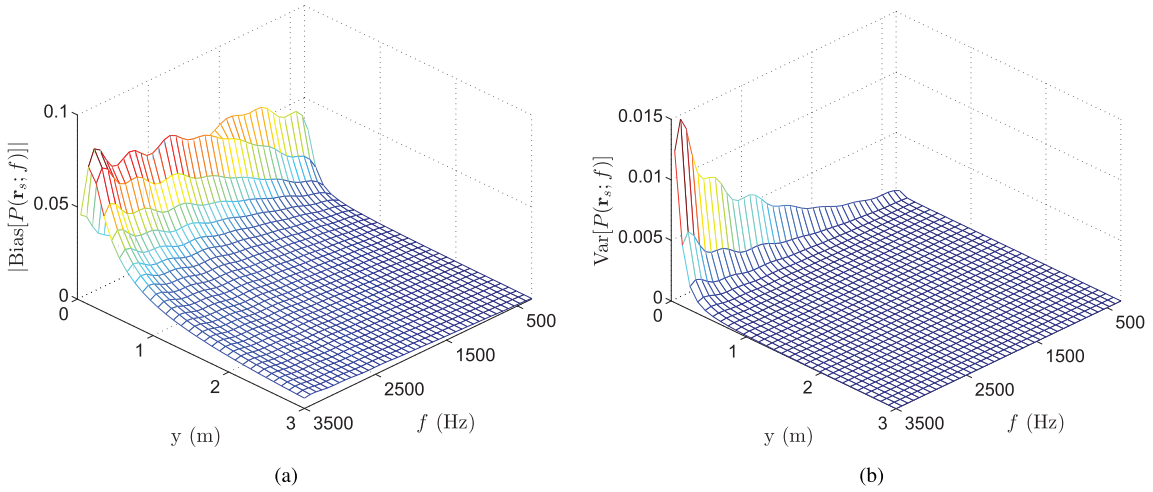


Fig. 6. The ARB and ARV of the nearfield wideband beamformer with different source-to-array distances in the presence of the omnidirectional perturbations in microphone positions. (a) The ARB. (b) The ARV.

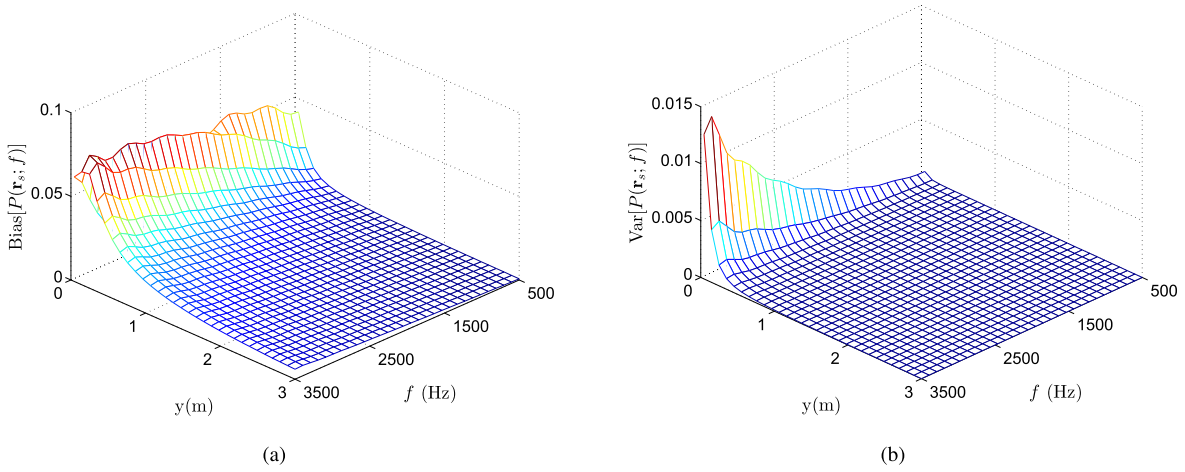


Fig. 7. The theoretical ARB and ARV of the nearfield wideband beamformer with different source-to-array distances in the presence of the omnidirectional perturbations in microphone positions. (a) The theoretical ARB, obtained by using (28). (b) The theoretical ARV, obtained by using (30).

order to reduce the effect of microphone position mismatches on farfield beamformers.

B. Nearfield Wideband Beamformers

Consider a nine-element uniform semi-circular array in nearfield with radius $r = 4.5$ cm. The nominal position of the mth microphone is $\mathbf{r}_m = [r\cos((m-1)\pi/8), r\sin((m-1)\pi/8), 0]^T$, $m = 0, \cdots, 8$. The FIR tap length is chosen as 30. The sample frequency is set to 8 kHz. The speed of sound is fixed at 340 m/s, unless otherwise stated. The desired passband response is chosen as $P_d(\mathbf{r}_s, f) = \exp\left[-j2\pi f\left(\frac{\|\mathbf{r}_s\|}{c} + \frac{L-1}{2f_s}\right)\right]$ while $P_d(\mathbf{r}_s, f) = 0$ in the stopband. The passband region is defined as $\Omega_P = \{(x, f)| -0.3 \le x \le 0.3, 300 \le f \le 3500\}$, and the stopband regions are $\Omega_{S_1} = \{(x, f)| -2.5 \le x \le -1, 0 \le f \le 4000\}$, $\Omega_{S_2} = \{(x, f)|1 \le x \le 2.5, 0 \le f \le 4000\}$, $\Omega_{S_3} = \{(x, f)| -1 \le x \le 1, 0 \le f \le 100\}$ and $\Omega_{S_4} = \{(x, f)| -1 \le x \le 1, 3700 \le f \le 4000\}$, respectively, where x is in meters and f in Hz. The designed beamformers are obtained by specifying on an x-axis parallel with, and 0.6 m in front of the array. The nearfield beamformer weights are normalized such that the peak of the mean array response magnitude is equal to unity to facilitate the performance comparison. All the results presented are obtained through 5000 Monte Carlo trials over a specified random set of microphone position perturbations.

1) Omnidirectional Perturbations in Microphone Positions: Suppose that the x-, y-, and z-axis perturbations in microphone positions all have a uniform distribution over $[-1, 2]$ mm. Figs. 6(a) and 6(b) show the ARB and ARV of the nearfield beamformer with the omnidirectional perturbations in microphone positions, respectively, where $x_s = 0.1$ m and $z_s = 0$. It can be seen that both the ARB and ARV will decrease when the source-to-array distance is increasing, which corroborates the theoretical findings that the robustness of nearfield wideband beamformers will be improved when the source is moving away from microphone arrays. For comparison, the theoretical ARB and ARV by (28) and (30) are also presented in Figs. 7(a) and 7(b), respectively. Again, it can be seen that the theoretical results are well-consistent with the simulation

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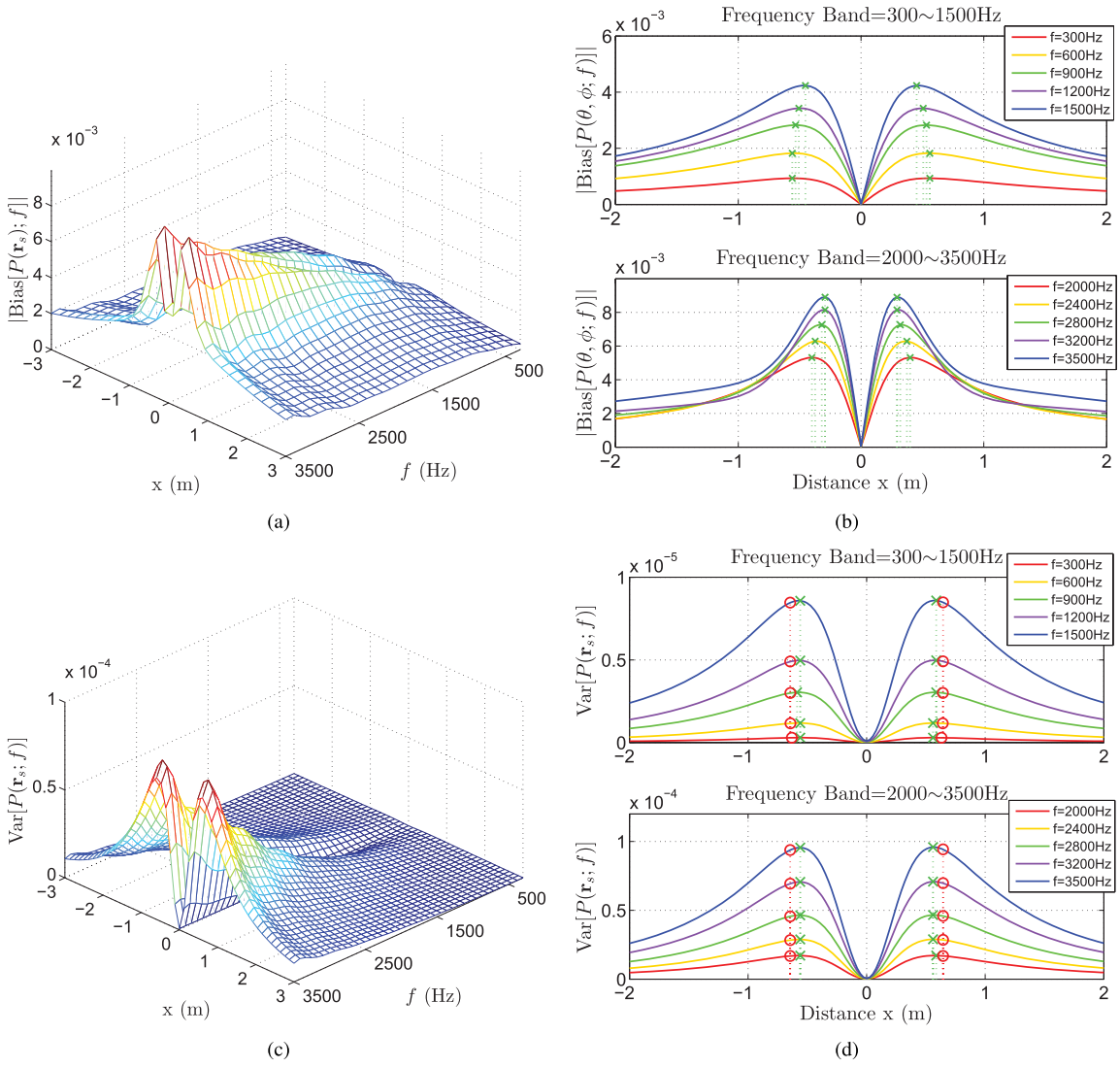


Fig. 8. The ARB and ARV of the nearfield wideband beamformer with different source-to-array distances along the x-axis in the presence of unidirectional perturbations in microphone positions. (a) The whole frequency band view of ARB. (b) The side views of ARB in different frequency bands, where the crosses represent the local peaks of each curve. (c) The whole frequency band view of ARV. (d) The side views of ARV in different frequency bands, where the circles denote the theoretical boundaries $\Delta x_{\min}$ and $\Delta x_{\max}$, and the crosses the local peaks of each curve.

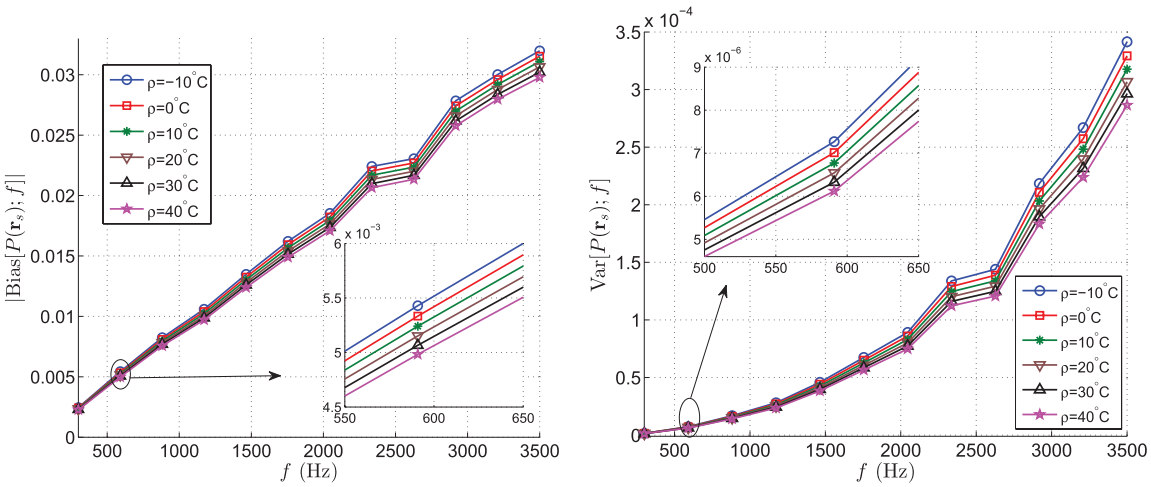
results in Figs. 6(a) and 6(b), though an approximation of nearfield array model has been employed in our previous analysis. It is also the case for the following examples. Due to space limit, we will not show the theoretical results in the following. However, it suffices to use the simulation results to support the above findings on nearfield beamformers.

2) Unidirectional Perturbations in Microphone Positions: Now we consider the case of unidirectional perturbation in microphone positions, where the perturbation is along the x-axis and uniformly distributed over $[-1, 2]$ mm. Fig. 8 displays the ARB and ARV of the nearfield beamformer with different source-to-array distances along the x-axis, where $y_s = 0.6$ m and $z_s = 0$. This is corresponding to the scenario that the perturbation direction of the microphone positions is parallel to the moving direction of the source.[11]

[11]The cases when the perturbation direction is perpendicular to source moving direction can also be similarly verified.

As shown in Figs. 8(a) and 8(b), the magnitude of ARB of the nearfield beamformer tends to decrease with the source-to-array distance increasing, which agrees with the above theoretical analysis. Figs. 8(c) and 8(d) show the ARV of the nearfield wideband beamformer. In Fig. 8(d), the boundaries of the uncertainty region, $\Delta x_{\min}$ and $\Delta x_{\max}$, marked by circles, are around -0.64 and 0.64 m, respectively. Note that the actual size of the uncertainty region is $0.64 \times 2 = 1.28$ m, which is very close to the theoretical value, $2r + 0.6 \times 2 = 1.29$ m, given in Finding 9.2. For ease of illustration, also shown in Fig. 8(d) are the local peaks for each curve marked by crosses, which are around -0.5 and 0.5 m, respectively. As can be seen clearly, outside the uncertainty region $(\Delta x_{\min}, \Delta x_{\max})$, the ARV of nearfield wideband beamformers is decreasing with the source-to-array distance increasing. Comparatively, there is no such uncertainty region for the case of omnidirectional perturbations in microphone positions as shown in Fig. 6(b). It is also interesting to note that, regarding the ARB, there also

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(a) (b)

Fig. 9. The ARB and ARV of the nearfield wideband beamformer with different temperatures in homogeneous environments. (a) The ARB. (b) The ARV, where the close-up views in lower frequency bands are shown in the insets.

exists an uncertainty region similar in size to the ARV. Outside the uncertainty region, the magnitude of ARB will decrease when the source is moving away from the microphone array.

3) Effect of Temperature in Homogeneous Environment: Finally, we consider the effect of temperature in homogeneous environment on nearfield wideband beamformers. Suppose that the x-, y-, and z-axis perturbations of microphone positions are all uniformly distributed over $[-1, 2]$ mm. Here the source location is set to $\mathbf{r}_s = (0.1, 0.6, 0)$ m. Fig. 9(b) shows the ARV of the nearfield beamformer with the temperature from -10 to 40 °C, i.e., the sound speed from 325.4 to 355.4 m/s. From the simulation results, it can be seen that the ARV of the nearfield beamformer will increase when the temperature is decreasing, i.e., when the speed of sound is decreasing. This is consistent well with our theoretical analysis. In addition, the ARB of the nearfield wideband beamformer is also shown in Fig. 9(a). Interestingly, similar to the ARV, the ARB will also increase with the temperature decreasing. Moreover, comparing with Figs. 5(a) and 5(b), we find that the effect of temperature on nearfield wideband beamformers is the same as that on their farfield counterparts.

VI. Conclusion

In this paper, the effects of sensor position errors on farfield/nearfield wideband beamformers for microphone arrays have been investigated. Some insights into the effects of sensor position errors on microphone arrays have been revealed through the comprehensive statistical analysis of farfield/nearfield wideband beamformers for arrays with arbitrary geometries.

In summary, for farfield wideband beamformers, the main findings are as follows:

- For widely used frequency-invariant beamformers, its array response will be more biased in high frequencies than in low frequencies.
- The ARB depends on source AoA. Also, the ARV depends on source AoA except the case with omnidirectional perturbations in microphone positions.
- For omnidirectional microphone position perturbations, the array response is nearly unbiased when source AoA is around $(\arccos(-\cot\phi/\sqrt{2}) + 45°, \phi)$ and $(\arccos(-\cot\phi/\sqrt{2}) - 135°, \phi)$ if $35.26° \leq \phi \leq 45°$, or around $(\arccos(-\cot\phi/\sqrt{2}) + 45°, \phi)$ and $(\arccos(-\cot\phi/\sqrt{2}) + 225°, \phi)$ if $45° < \phi \leq 144.73°$. However, the source AoA has little effect on the ARV. Consequently, when source AoA is around the directions specified above, the farfield wideband beamformers will be more robust.
- For unidirectional microphone position perturbations, the magnitude of ARB is proportional to the magnitude of the cosine of the angle between the direction vector of the source and that of microphone position perturbations, while the ARV is proportional to the square of the cosine of the angle between the two vectors. Therefore, farfield wideband beamformers will be more robust when source AoA is perpendicular to the perturbation direction of sensor positions.
- The ARV will increase with the speed of sound decreasing, i.e., with the temperature decreasing in homogeneous environment. This is also the case for the ARB, if the nominal array response is relatively insensitive to variation of the speed of sound. Accordingly, the farfield wideband beamformers are more robust at higher temperatures in the presence of sensor position errors.

Regarding nearfield wideband beamformers, the main findings are summarized as follows:

- The ARB will tend towards zero as the source-to-array distance increases, regardless of unidirectional or omnidirectional perturbations in microphone positions.
- For omnidirectional microphone position perturbations, the ARV will increase when the source-to-array distance is decreasing. Considering the ARB property, it is concluded that the robustness of nearfield wideband beamformers tends to be improved when the source is moving away from microphone array.
- For unidirectional microphone position perturbations, the ARV will decrease when the source moves away from the

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arrays outside an uncertainty region whose boundaries depend on microphone locations when the microphone perturbation direction is perpendicular to the source direction, or both the source and microphone locations when the microphone perturbation direction is parallel to the source direction. Reconsidering the ARB property, it follows that the robustness of a nearfield wideband beamformer will be improved with the source-to-array distance increasing outside the uncertainty region.

- Unlike the effect of microphone gain and phase errors, in the presence of microphone position errors the ARV of a nearfield wideband beamformer will increase with the speed of sound decreasing, i.e., with the temperature increasing in homogeneous environments.

Simulations results have been presented to corroborate the theoretical analysis. These contributions are helpful to better understand the robustness properties of microphone arrays in the presence of sensor position errors and also useful in microphone array design.

APPENDIX A
PROOF OF FINDING 2

If $\text{Bias}[\widetilde{P}(\theta,\phi;\,f)]=0$, then by (20) we have $\mathbf{u}_s^T\boldsymbol{\mu}=0$. Recall that $\mu_x=\mu_y=\mu_z$ for omnidirectional perturbations in microphone positions. Therefore, it follows that

$$\sin\phi\cos\theta+\sin\phi\sin\theta+\cos\phi=0. \tag{37}$$

Eq. (37) can be reformulated as

$$\cos(\theta-45^\circ)=-\cot\phi/\sqrt{2}. \tag{38}$$

Note that $|\cos(\theta-45^\circ)|\le 1$ and $\phi\in[0,180^\circ]$, thus we have $|\cot\phi|\le\sqrt{2}$, which implies $35.26^\circ\le\phi\le 144.73^\circ$. Considering $\theta\in[0,360^\circ)$, by (38) we can conclude that the array response will be nearly unbiased when the source AOA is around $(\arccos(-\cot\phi/\sqrt{2})+45^\circ,\phi)$ and $(\arccos(-\cot\phi/\sqrt{2})-135^\circ,\phi)$ if $35.26^\circ\le\phi\le 45^\circ$, or around $(\arccos(-\cot\phi/\sqrt{2})+45^\circ,\phi)$ and $(\arccos(-\cot\phi/\sqrt{2})+225^\circ,\phi)$ if $45^\circ<\phi\le 144.73^\circ$.

APPENDIX B
PROOF OF THEOREM 2

With (14), we have

$$\begin{aligned}
&\text{Var}[\widetilde{P}(\theta,\phi;\,f)]\\
&\quad=\text{Var}[\mathbf{w}^T\Delta\mathbf{a}(\theta,\phi;\,f)]\\
&\quad=\mathbf{w}^T\mathbb{E}\big\{(\Delta\mathbf{a}(\theta,\phi;\,f)-\mathbb{E}[\Delta\mathbf{a}(\theta,\phi;\,f)])\\
&\qquad\times(\Delta\mathbf{a}^H(\theta,\phi;\,f)-\mathbb{E}[\Delta\mathbf{a}^H(\theta,\phi;\,f)])\big\}\mathbf{w}\\
&\quad=\mathbf{w}^T\big\{\mathbb{E}[\Delta\mathbf{a}(\theta,\phi;\,f)\Delta\mathbf{a}^H(\theta,\phi;\,f)]\\
&\qquad-\mathbb{E}[\Delta\mathbf{a}(\theta,\phi;\,f)]\mathbb{E}[\Delta\mathbf{a}^H(\theta,\phi;\,f)]\big\}\mathbf{w}\\
&\quad=\mathbf{w}^T\mathbf{Q}(\theta,\phi;\,f)\mathbf{w}
\end{aligned}\tag{39}$$

where the superscript $(\cdot)^H$ represents the Hermitian transpose, and

$$\begin{aligned}
\mathbf{Q}(\theta,\phi;\,f)\triangleq\;&\mathbb{E}[\Delta\mathbf{a}(\theta,\phi;\,f)\Delta\mathbf{a}^H(\theta,\phi;\,f)]\\
&-\mathbb{E}[\Delta\mathbf{a}(\theta,\phi;\,f)]\mathbb{E}[\Delta\mathbf{a}^H(\theta,\phi;\,f)].
\end{aligned}\tag{40}$$

Using (3), (15) and (18), we can represent the ith element of $\Delta\mathbf{a}(\theta,\phi;\,f)$ as

$$\begin{aligned}
\Delta a_i(\theta,\phi;\,f)\simeq\;&(j2\pi f\mathbf{u}_s^T\Delta\mathbf{r}_m/c)\\
&\times\exp\left\{-j2\pi f(k/f_s+\mathbf{u}_s^T\mathbf{r}_m/c)\right\}
\end{aligned}\tag{41}$$

with $k=\ \bmod(i-1,L)$, $m=\lfloor(i-1)/L\rfloor$, $i=1,2,\ldots,ML$, where $\bmod(i-1,L)$ is the remainder of $(i-1)/L$, and $\lfloor(i-1)/L\rfloor$ denotes the largest integer smaller than or equal to $(i-1)/L$.

Substituting (41) into (40), we can obtain the (i,j)th element of $\mathbf{Q}(\theta,\phi;\,f)$ as

$$\begin{aligned}
&\mathbf{Q}^{(ij)}(\theta,\phi;\,f)\\
&=\mathbb{E}[\Delta a_i(\theta,\phi;\,f)\Delta a_j^*(\theta,\phi;\,f)]\\
&\quad-\mathbb{E}[\Delta a_i(\theta,\phi;\,f)]\mathbb{E}[\Delta a_j^*(\theta,\phi;\,f)]\\
&\simeq(2\pi f/c)^2\Big\{(\sin\phi\cos\theta)^2\left(\mathbb{E}[\delta_m^x\delta_n^x]-\mathbb{E}[\delta_m^x]\mathbb{E}[\delta_n^x]\right)\\
&\qquad+(\sin\phi\sin\theta)^2\left(\mathbb{E}[\delta_m^y\delta_n^y]-\mathbb{E}[\delta_m^y]\mathbb{E}[\delta_n^y]\right)\\
&\qquad+(\cos\phi)^2\left(\mathbb{E}[\delta_m^z\delta_n^z]-\mathbb{E}[\delta_m^z]\mathbb{E}[\delta_n^z]\right)\Big\}\\
&\quad\times\exp\left\{-j2\pi f\left[(k-l)/f_s+\mathbf{u}_s^T(\mathbf{r}_m-\mathbf{r}_n)/c\right]\right\}\\
&=\begin{cases}(2\pi f/c)^2[\sigma_x^2(\sin\phi\cos\theta)^2+\sigma_y^2(\sin\phi\sin\theta)^2+\sigma_z^2\\
\times(\cos\phi)^2]\exp\{-j2\pi(k-l)f/f_s\},&\text{if } n=m,\\
0,&\text{otherwise}\end{cases}
\end{aligned}\tag{42}$$

where the superscript $(\cdot)^*$ denotes the complex conjugate, and $k=\ \bmod(i-1,L)$, $l=\ \bmod(j-1,L)$, $m=\lfloor(i-1)/L\rfloor$, $n=\lfloor(j-1)/L\rfloor$.

From (42), we know the imaginary part of $\mathbf{Q}(\theta,\phi;\,f)$, denoted as $\text{Im}\{\mathbf{Q}(\theta,\phi;\,f)\}$, is skew-symmetric. Therefore, it follows that $\mathbf{w}^T\text{Im}\{\mathbf{Q}(\theta,\phi;\,f)\}\mathbf{w}=0$. Consequently, we have

$$\begin{aligned}
&\text{Var}[\widetilde{P}(\theta,\phi;\,f)]\\
&\quad=\mathbf{w}^T[\text{Re}\{\mathbf{Q}(\theta,\phi;\,f)\}+j\text{Im}\{\mathbf{Q}(\theta,\phi;\,f)\}]\mathbf{w}\\
&\quad\simeq\mathbf{w}^T\text{Re}\{\mathbf{Q}(\theta,\phi;\,f)\}\mathbf{w}\\
&\quad=(2\pi f/c)^2[\sigma_x^2(\sin\phi\cos\theta)^2+\sigma_y^2(\sin\phi\sin\theta)^2\\
&\qquad+\sigma_z^2(\cos\phi)^2]\mathbf{w}^T[\mathbf{I}_M\otimes\Lambda(f)]\mathbf{w}
\end{aligned}$$

where $\text{Re}\{\cdot\}$ denotes the real part.

APPENDIX C
PROOF OF PROPOSITION 3

To prove Proposition 3, we need to introduce the following lemmas.

Lemma 1: Consider the function $f(t)=\frac{(t-\alpha)^2}{[(t-\alpha)^2+\beta]^3}+\frac{\gamma(t-\alpha)^2}{[(t-\alpha)^2+\beta]^2}$, where $t,\alpha\in\mathbb{R}$, and $\beta,\gamma>0$. It holds that $f(t)$ is increasing in both $\left(-\infty,\alpha-\sqrt{\frac{\sqrt{\beta\gamma(\beta\gamma+1)}+1-1}{\gamma}}\right]$ and $\left(\alpha,\alpha+\sqrt{\frac{\sqrt{\beta\gamma(\beta\gamma+1)}+1-1}{\gamma}}\right)$, and is decreasing in both $\left(\alpha-\sqrt{\frac{\sqrt{\beta\gamma(\beta\gamma+1)}+1-1}{\gamma}},\alpha\right]$ and $\left[\alpha+\sqrt{\frac{\sqrt{\beta\gamma(\beta\gamma+1)}+1-1}{\gamma}},+\infty\right)$.

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Proof: The derivation of $f(t)$ is given by

$$\begin{aligned} & f'(t) \\ &= \frac{2(t-\alpha)}{[(t-\alpha)^2+\beta]^3} - \frac{6(t-\alpha)^3}{[(t-\alpha)^2+\beta]^4} + \gamma\,\frac{2\gamma\,(t-\alpha)}{[(t-\alpha)^2+\beta]^2} \\ &\quad - \frac{4\gamma\,(t-\alpha)^3}{[(t-\alpha)^2+\beta]^3} \\ &= \frac{2(t-\alpha)}{[(t-\alpha)^2+\beta]^4}[-\gamma\,(t-\alpha)^4 - 2(t-\alpha)^2 + \beta^2\gamma\, + \beta] \\ &= \frac{2(t-\alpha)}{[(t-\alpha)^2+\beta]^4}\left\{\left(t-\alpha+\sqrt{\frac{\sqrt{\beta\gamma\,(\beta\gamma\,+1)+1}-1}{\gamma}}\right)\right. \\ &\qquad \times \left(t-\alpha-\sqrt{\frac{\sqrt{\beta\gamma\,(\beta\gamma\,+1)+1}-1}{\gamma}}\right) \\ &\qquad \left.\times\left[(t-\alpha)^2+\frac{\sqrt{\beta\gamma\,(\beta\gamma\,+1)+1}-1}{\gamma}\right]\right\} \end{aligned} \tag{43}$$

Since $\beta, \gamma > 0$, we obtain $\sqrt{\beta\gamma\,(\beta\gamma+1)+1} > 1$, which implies $\frac{\sqrt{\beta\gamma\,(\beta\gamma+1)+1}-1}{\gamma} > 0$. Using Eq. (43), we have $f'(t) \geq 0$ in $\left(-\infty, \alpha - \sqrt{\frac{\sqrt{\beta\gamma\,(\beta\gamma+1)+1}-1}{\gamma}}\right]$ and $\left(\alpha, \alpha+\sqrt{\frac{\sqrt{\beta\gamma\,(\beta\gamma+1)+1}-1}{\gamma}}\right)$. Therefore, $f(t)$ is increasing in these intervals. Since $f'(t) \leq 0$, in $\left(\alpha-\sqrt{\frac{\sqrt{\beta\gamma\,(\beta\gamma+1)+1}-1}{\gamma}}, \alpha\right]$ and $\left[\alpha+\sqrt{\frac{\sqrt{\beta\gamma\,(\beta\gamma+1)+1}-1}{\gamma}}, +\infty\right)$, it follows that $f(t)$ is decreasing in these intervals. ■

Lemma 2: Consider the function $f(t) = \frac{\varepsilon}{[(t-\alpha)^2+\beta]^3} + \frac{\varepsilon\gamma}{[(t-\alpha)^2+\beta]^2}$, where $t, \alpha \in \mathbb{R}$, and $\varepsilon, \beta, \gamma > 0$. It holds that $f(t)$ is increasing in $t \in (-\infty, \alpha]$, and decreasing in $t \in (\alpha, +\infty)$.

Proof: The derivation of $f(t)$ is given by

$$f'(t) = -2\varepsilon(t-\alpha)\left(\frac{3}{[(t-\alpha)^2+\beta]^4} + \frac{2\gamma}{[(t-\alpha)^2+\beta]^3}\right). \tag{44}$$

With Eq. (44), we have $f'(t) \geq 0$ in $(-\infty, \alpha]$ and $f'(t) < 0$ in $(\alpha, +\infty)$. Therefore, $f(t)$ is increasing in $t \in (-\infty, \alpha]$, and decreasing in $t \in (\alpha, +\infty)$. ■

Lemma 3: For two matrices $\mathbf{A}$ and $\mathbf{B}$, it holds that [33]: 1) if $\mathbf{A}$ and $\mathbf{A} \otimes \mathbf{B}$ are both positive semi-definite, then $\mathbf{B}$ is also positive semi-definite; 2) if $\mathbf{A}$ and $\mathbf{B}$ are both positive semi-definite, then $\mathbf{A} \otimes \mathbf{B}$ is also positive semi-definite.

Firstly, we give the proof of Proposition 3.1. Let $t = x_s$, $\alpha = x_m, \beta = (y_s - y_m)^2 + (z_s - z_m)^2$, $\gamma = (2\pi f/c)^2$, and define $f(t) \triangleq \Phi_x^{(mm)}(x_s, y_s, z_s; f)$. From Lemma 1, it follows that $\Phi_x^{(mm)}(x_s, y_s, z_s; f)$ is increasing in $(-\infty, x_m - X_m]$, and decreasing in $[x_m + X_m, +\infty)$, where X_m is defined in (36). Therefore, for $x_1 \leq x_2 \leq \Delta x_{\min}$ or $x_1 \geq x_2 \geq \Delta x_{\max}$, where $\Delta x_{\min} = \min\{x_m - X_m\}$, $\Delta x_{\max} = \max\{x_m + X_m\}$, we have $\Phi_x^{(mm)}(x_1, y_s, z_s; f) \leq \Phi_x^{(mm)}(x_2, y_s, z_s; f)$ for $\forall m$. This implies the matrix $\Phi_x(x_2, y_s, z_s; f) - \Phi_x(x_1, y_s, z_s; f)$ is positive semi-definite. Considering the nonnegativity of (30), we know that $\Phi_x(\mathbf{r}, f) \otimes \Lambda(f)$ is positive semi-definite. As shown in (31), the matrix $\Phi_x(\mathbf{r}, f)$ is positive semi-definite. Using Lemma 1, we obtain the matrix $\Lambda(f)$ is also positive semi-definite. Therefore, using Lemma 2, we have $\mathrm{Var}[\widetilde{P}(x_2, y_s, z_s; f)] - \mathrm{Var}[\widetilde{P}(x_1, y_s, z_s; f)] \simeq \sigma_x^2 \mathbf{w}^T\{[\Phi_x(x_2, y_s, z_s; f) - \Phi_x(x_1, y_s, z_s; f)] \otimes \Lambda(f)\}\mathbf{w} \geq 0$, i.e., $\mathrm{Var}[\widetilde{P}(x_1, y_s, z_s; f)] \leq \mathrm{Var}[\widetilde{P}(x_2, y_s, z_s; f)]$.

Next, we prove Proposition 3.2. Let $t = y_s$, $\alpha = y_m$, $\varepsilon = x_s^2$, $\beta = (x_s - x_m)^2 + (z_s - z_m)^2$ and $\gamma = (2\pi f/c)^2$. Using Lemma 2, it follows that $f(t) \triangleq \Phi_x^{(mm)}(\mathbf{r}_s; f)$ and $\Phi^{(mm)}(x_s, y_s, z_s; f)$ are both increasing in $(-\infty, y_m]$ and both decreasing in $(y_m, +\infty)$. Thus, for $y_1 \leq y_2 \leq y_{\min}$ or $y_1 \geq y_2 \geq y_{\max}$, where $y_{\min} = \min\{y_m\}$, $y_{\max} = \max\{y_m\}$, it holds that $\Phi_x^{(mm)}(x_s, y_1, z_s; f) \leq \Phi_x^{(mm)}(x_2, y_2, z_s; f)$, $\forall m$. Reconsidering the positive semi-definiteness of the matrix $\Lambda(f)$ and using Lemma 2, we can conclude that $\mathrm{Var}[\widetilde{P}(x_s, y_1, z_s; f)] \leq \mathrm{Var}[\widetilde{P}(x_s, y_2, z_s; f)]$. Following the similar procedures, Proposition 3.3 can be proved.

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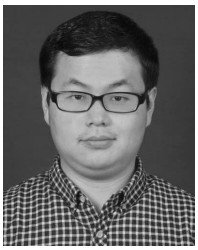
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